



The World's Ontology Ecosystem

Day 1

INSTRUCTOR



John Beverley, PhD

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INSTRUCTOR



Barry Smith, PhD

Chairman of Board, *National Center for Ontological
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SUNY Distinguished Professor of Philosophy
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Welcome

- Logistics
- Agenda
- Goals of the Course

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Online Participation

- There are **~500 people** registered to attend online
- We will be recording these sessions to be posted online for asynchronous viewing after the event
- We will also be uploading any lectures to the course GitHub page:

<https://github.com/Applied-Ontology-Education/World-s-Ontology-Ecosystem>

Questions

- Please keep yourself muted unless asking a question; when the time comes for questions, please use the “raise hand” feature to form a queue
- We will be monitoring the chat if there are technical questions that need addressing
- Questions regarding content should be held until allotted times at the ends of presentations or posted in the GitHub discussion forum:

<https://github.com/Applied-Ontology-Education/World-s-Ontology-Ecosystem/discussions>

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Time	Session	Lead	Focus and participant output
9:00–9:15	Welcome	John Beverley	Logistics, introductions, course goals, and an overview of the week.
9:15–10:15	History of Ontology	Barry Smith (remote)	From ontology’s philosophical origins to modern data modeling.
10:15–10:30	Break		
10:30–11:30	The World’s Ontology Ecosystem	Barry Smith (remote)	The landscape of modern ontology efforts within and beyond the Basic Formal Ontology ecosystem.
11:30–12:00	Ontology Foundry Efforts	John Beverley	Foundry initiatives that apply ontology engineering best practices, including the Industrial Ontologies Foundry and the National Security Ontology Federation.
12:00–1:00	Lunch		
1:00–2:00	Working Ontologists: Air Force	Denise Ferniza	The importance of applying ontologies in real-world systems and lessons learned from field practice.
2:00–3:00	Ontology Tradecraft	John Beverley	Orientation to the Semantic Web toolkit, ontology engineering best practices, and strategies for evaluating designs.
3:00–3:15	Break		
3:15–4:45	Exercise	John Beverley and Denise Ferniza	Build preliminary models with semantic stack and best practices Open Day 1 exercise materials
4:45–5:00	Closing Remarks	John Beverley	Review the day’s central lessons and preview Day 2.

<https://ncor-network.org/summer-school-2026>

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Course Objectives

- **Day 1:** Orient participants to applied ontology, why ontologies are developed, and the tradecraft used throughout the week
- **Day 2:** Teach a repeatable modeling patterns and apply across challenging domains
- **Day 3:** With healthy skepticism, show how ontologies ground AI systems, and provide practical guidance for using LLMs in ontology engineering
- **Day 4:** Examine applied research, ontology-driven agents, semantic pipelines, and methods for validating ontology-enabled AI workflows
- **Day 5:** Synthesize lessons through competing visions for the future of the field